

Wangyang Ying

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Research Interests

My research interests lie in **data-centric AI**, **multi-agent reasoning**, and **scientific equation discovery**. In particular, I focus on data-centric methods to enhance the robustness and effectiveness of machine learning; multi-agent frameworks for structured knowledge extraction and reasoning; and interpretable methods for equation discovery to uncover scientific patterns from data.

Education

Ph.D., Computer Science, Arizona State University, 2023 - present
M.S., Computer Science, Sichuan University, 2016 - 2019
B.E., Computer Science, Sichuan University, 2012 - 2016

Work Experience

Research Intern, Data Science & System Security Department
NEC Laboratories America, Inc., Princeton, May 2025 - August 2025

- Developed **multi-agent LLM** frameworks for structured knowledge extraction (procedural graph representation), supporting downstream retrieval-augmented generation (RAG). Explored how structured knowledge enables **personalized LLM** training by grounding user-specific workflows into structured representations.

Research Intern, Institute of High Performance Computing
A*STAR, Singapore, June 2024 - August 2024

- Investigated **trustworthiness of LLMs** in medical applications, with emphasis on understanding how **jail-break attacks** compromise system reliability. Conducted systematic analysis of jailbreak strategies as a foundation for designing future LLM safety and protection mechanisms.

Full Time, Platform and Content Group
Tencent, Beijing, November 2020 - August 2022

- Work for **time-sensitive search scenarios** (e.g., weather, stock, news), serving hundreds of millions of users. Design algorithms for query time-sensitivity detection, retrieval pipeline optimization, and time-aware ranking and presentation to enhance freshness and relevance in search results.

Full Time, Digital Media & Entertainment Group
Alibaba, Beijing, June 2019 - October 2020

- Built **recommendation systems** for long- and short-form video platforms (movies, TV shows, variety shows, and micro-videos). Worked on **video content understanding** (e.g., tagging, user profiling) and **video retrieval**, improving large-scale recommendation quality and user engagement.

Publications

Refereed Journal Publications

*: Denoted equal contributions.

1. [TKDD] **Wangyang Ying**, Haoyue Bai, Kunpeng Liu, Yanjie Fu. Topology-aware Reinforcement Feature Space Reconstruction for Graph Data. ACM Transactions on Knowledge Discovery from Data, 2025.
2. [TKDD] **Wangyang Ying**, Dongjie Wang, Haifeng Chen, Yanjie Fu. Feature Selection as Deep Sequential Generative Learning. ACM Transactions on Knowledge Discovery from Data, 2024.

3. [TIST] Nanxu Gong*, **Wangyang Ying***, Dongjie Wang, Yanjie Fu. Neuro-Symbolic Embedding for Short and Effective Feature Selection via Autoregressive Generation. ACM Transactions on Intelligent Systems and Technology, 2024.
4. [npj Artificial Intelligence] Haoyue Bai, **Wangyang Ying**, Nanxu Gong, Xinyuan Wang, Hao Liu, Yanjie Fu. Privacy Preserving Generative Feature Transformation, 2025.
5. [TBD] Xinyuan Wang, Dongjie Wang, **Wangyang Ying**, Rui Xie, Haifeng Chen, Yanjie Fu. Knockoff-Guided Feature Selection via A Single Pre-trained Reinforced Agent. IEEE Transactions on Big Data, 2025.
6. [FCS] **Wangyang Ying**, Lei Zhang, Hongli Deng. Sichuan dialect speech recognition with deep LSTM network. Frontiers of Computer Science, 2020.

Conference Proceedings

1. [KDD] **Wangyang Ying**, Dongjie Wang, Xuanming Hu, Yuanchun Zhou, Charu C. Aggarwal, Yanjie Fu. Unsupervised Generative Feature Transformation via Graph Contrastive Pre-training and Multi-objective Fine-tuning. Proceedings of the 30th ACM SIGKDD Conference on Knowledge Discovery and Data Mining, 2024.
2. [CIKM] **Wangyang Ying**, Dongjie Wang, Xuanming Hu, Ji Qiu, Jin Park, Yanjie Fu. Revolutionizing Biomarker Discovery: Leveraging Generative AI for Bio-Knowledge-Embedded Continuous Space Exploration. In Proceedings of the 33rd ACM International Conference on Information and Knowledge Management, 2024.
3. [ICDM] **Wangyang Ying**, Dongjie Wang, Kunpeng Liu, Leilei Sun, Yanjie Fu. Self-Optimizing Feature Generation via Categorical Hashing Representation and Hierarchical Reinforcement Crossing. The 23rd IEEE International Conference on Data Mining, 2023.
4. [AAAI] Xinyuan Wang, Dongjie Wang, **Wangyang Ying**, Haoyue Bai, Nanxu Gong, Sixun Dong, Kunpeng Liu, Yanjie Fu. Efficient Post-Training Refinement of Latent Reasoning in Large Language Models, The Association for the Advancement of Artificial Intelligence, 2026.
5. [AAAI] Haoyue Bai, Guodong Chen, **Wangyang Ying**, Xinyuan Wang, Nanxu Gong, Sixun Dong, Giulia Pedrielli, Haoyu Wang, Haifeng Chen, Yanjie Fu. Brownian Bridge Augmented Surrogate Simulation and Injection Planning for Geological CO₂ Storage, The Association for the Advancement of Artificial Intelligence, 2026.
6. [NeurIPS] Nanxu Gong, Zijun Li, Sixun Dong, Haoyue Bai, **Wangyang Ying**, Xinyuan Wang, Yanjie Fu. Sculpting Features from Noise: Reward-Guided Hierarchical Diffusion for Task-Optimal Feature Transformation. The 39th Annual Conference on Neural Information Processing Systems, 2025.
7. [IJCAI] Nanxu Gong, Xinyuan Wang, **Wangyang Ying**, Haoyue Bai, Sixun Dong, Haifeng Chen, Yanjie Fu. Unsupervised Feature Transformation via In-context Generation, Generator-critic LLM Agents, and Duet-play Teaming. International Joint Conference on Artificial Intelligence, 2025.
8. [AAAI] Nanxu Gong, Chandan K. Reddy, **Wangyang Ying**, Yanjie Fu. Evolutionary Large Language Model for Automated Feature Transformation. The Association for the Advancement of Artificial Intelligence, 2025.
9. [CIKM] Xuanming Hu, Dongjie Wang, **Wangyang Ying**, Yanjie Fu. Reinforcement Feature Transformation for Polymer Property Performance Prediction. In Proceedings of the 33rd ACM International Conference on Information and Knowledge Management, 2024.
10. [EMNLP] Haolin Deng, Yanan Zhang, Yangfan Zhang, **Wangyang Ying**, Changlong Yu, Jun Gao, Wei Wang, Xiaoling Bai, Nan Yang, Jin Ma, Xiang Chen, Tianhua Zhou. Title2Event: Benchmarking Open Event Extraction with a Large-scale Chinese Title Dataset. The 2022 Conference on Empirical Methods in Natural Language Processing, 2022.

11. [WSC] Haoyue Bai, Haoyu Wang, Nanxu Gong, Xinyuan Wang, **Wangyang Ying**, Haifeng Chen, Yanjie Fu. Supply Chain Optimization via Generative Simulation and Iterative Decision Policies. Winter Simulation Conference, 2025.

Preprints

1. [Arxiv] **Wangyang Ying**, Jinghan Zhang, Haoyue Bai, Nanxu Gong, Xinyuan Wang, Kunpeng Liu, Chandan K Reddy, Yanjie Fu. Data-Efficient Symbolic Regression via Foundation Model Distillation, 2025.
2. [Arxiv] **Wangyang Ying**, Haoyue Bai, Nanxu Gong, Xinyuan Wang, Sixun Dong, Haifeng Chen, Yanjie Fu. Bridging the Domain Gap in Equation Distillation with Reinforcement Feedback, 2025.
3. [Arxiv] **Wangyang Ying**, Nanxu Gong, Dongjie Wang, Xinyuan Wang, Arun Vignesh Malarkkan, Vivek Gupta, Chandan K Reddy, Yanjie Fu. Distribution Shift Aware Neural Tabular Learning, 2025.
4. [Arxiv] **Wangyang Ying**, Cong Wei, Nanxu Gong, Xinyuan Wang, Haoyue Bai, Arun Vignesh Malarkkan, Sixun Dong, Dongjie Wang, Denghui Zhang, Yanjie Fu. A Survey on Data-Centric AI: Tabular Learning from Reinforcement Learning and Generative AI Perspective, 2025.
5. [Arxiv] Xinyuan Wang, Haoyue Bai, Nanxu Gong, **Wangyang Ying**, Sixun Dong, Xiquan Cui, Yanjie Fu. LLM-ML Teaming: Integrated Symbolic Decoding and Gradient Search for Valid and Stable Generative Feature Transformation, 2025.
6. [Arxiv] Nanxu Gong, Sixun Dong, Haoyue Bai, Xinyuan Wang, **Wangyang Ying**, Yanjie Fu. Agentic Feature Augmentation: Unifying Selection and Generation with Teaming, Planning, and Memories, 2025.

Honors

- KDD 2024 Travel Award
- National Scholarship, Top 1%, 2019
- Postgraduate Scholarship in Sichuan University, 1st Prize, Top 1%, 2018, 2019
- Outstanding volunteer of the 8th China Intelligence Industry Summit, 2018
- Robomaster Competition, 3rd Prize, 2015

Professional Activities and Services

- Neural Information Processing Systems (**NuerIPS**) 2025
- International Conference on Machine Learning (**ICML**) 2025
- The International Conference on Learning Representations (**ICLR**) 2024, 2025, 2026
- The Conference on Knowledge Discovery and Data Mining (**KDD**) 2024, 2025
- The Association for the Advancement of Artificial Intelligence (**AAAI**) 2025
- The Conference on Information and Knowledge Management (**CIKM**) 2023, 2024
- The IEEE International Conference on Big Data (**BigData**) 2023, 2024, 2025
- ACM Transactions on Knowledge Discovery from Data (**TKDD**)
- IEEE Transactions on Knowledge and Data Engineering (**TKDE**)
- IEEE Transactions on Cognitive and Developmental Systems (**TCDS**)
- IEEE Transactions on Big Data (**TBD**)